

ARYAMAN PANIGRAHI

Computer Science Engineering (Cybersecurity) Student

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SUMMARY

Computer Science undergraduate at VIT Vellore who builds software across the stack — from a deployed full-stack document-management platform (FastAPI, PostgreSQL, React) to signal-processing pipelines in Python — with a specialization in Cybersecurity and hands-on experience evaluating LLM safety through adversarial testing. Grounded in Data Structures & Algorithms and Object-Oriented Programming, with 100+ solved LeetCode problems and a continuous-learning habit spanning software engineering, AI/ML, and applied security.

EXPERIENCE

Tech Mahindra — Cybersecurity & Quantum Computing Intern Summer 2026 (8 Weeks)

- Designed and executed 100+ adversarial test scenarios against Large Language Models — including multi-step prompt injection, authority spoofing, and poisoned legal-reference attacks — to evaluate model safety and robustness.
- Proposed 5 additional adversarial scenarios that were incorporated into the evaluation suite, extending test coverage for guardrail validation.
- Assisted in implementing and validating AI guardrails, contributing to a more resilient LLM safety-testing workflow.
- Developed basic quantum circuits in Qiskit as part of an internal assignment, applying quantum gates, superposition, and entanglement concepts using Python.

PROJECTS

DocVault — Full-Stack Document Management Platform Python · FastAPI · React

- Designed a client-agnostic FastAPI backend with JWT authentication (bcrypt), PostgreSQL via SQLAlchemy + Alembic migrations, and S3-compatible object storage (MinIO in dev, Backblaze B2 in production).
- Built automatic OCR extraction (Tesseract + PyMuPDF) powering full-text search, plus soft-delete/restore, sorting, and drag-and-drop upload.
- Built the React, TypeScript & Tailwind CSS frontend and deployed end-to-end (Vercel, Render, Neon). Live: doc-vault-jet-kappa.vercel.app

Acoustic Source Localization & Classification Python

- Built a Python-based simulation using a 4-microphone array to estimate sound source direction via Time Difference of Arrival (TDOA) algorithms, achieving 0–23° angular error (≈11° average) across test scenarios in simulated testing.
- Engineered a modular signal-processing pipeline — signal generation, TDOA localization, and sound classification — with a visualization layer for direction-of-arrival results.

Disaster Alert Dashboard React

- Designed and deployed a responsive front-end dashboard in React to surface disaster alerts and safety information, focusing on component structure and clean UI/UX presentation.
- Live: disaster-alert-dashboard-xi.vercel.app

TECHNICAL SKILLS

Languages: Python, C++, C, Java, TypeScript

Frameworks & Libraries: React, FastAPI, SQLAlchemy, Tailwind CSS, Qiskit

Developer Tools: Git, GitHub, Docker, VS Code

Databases & Cloud: PostgreSQL, MinIO / Backblaze B2 (Object Storage), Render, Vercel, Neon

Cybersecurity: Adversarial prompt testing, LLM guardrail validation, security risk management, AI safety evaluation, JWT authentication (bcrypt)

AI/ML: LLM evaluation, adversarial prompt engineering, quantum computing fundamentals (Qiskit)

Concepts: Data Structures & Algorithms, Object-Oriented Programming, Signal Processing, REST API Design, Problem Solving

EDUCATION

VIT University, Vellore — B.Tech, Computer Science Engineering (Cybersecurity) 2024–2028

CGPA: 8.85 · JEE Mains: 91st Percentile

CERTIFICATIONS

Google Cybersecurity Professional Certificate — 2 of 9 courses completed

- [Foundations of Cybersecurity](#)
- [Play It Safe: Manage Security Risks](#)

ACHIEVEMENTS

- Solved 100+ problems on LeetCode, strengthening algorithmic problem-solving and coding proficiency.
- Scored in the 91st percentile in JEE Mains.
- Proposed 5 new adversarial test scenarios adopted into Tech Mahindra's LLM evaluation suite.

EXTRA-CURRICULAR

- Mozilla Firefox Club, VIT — Junior Core Member:** active member contributing to club initiatives and technical community engagement.